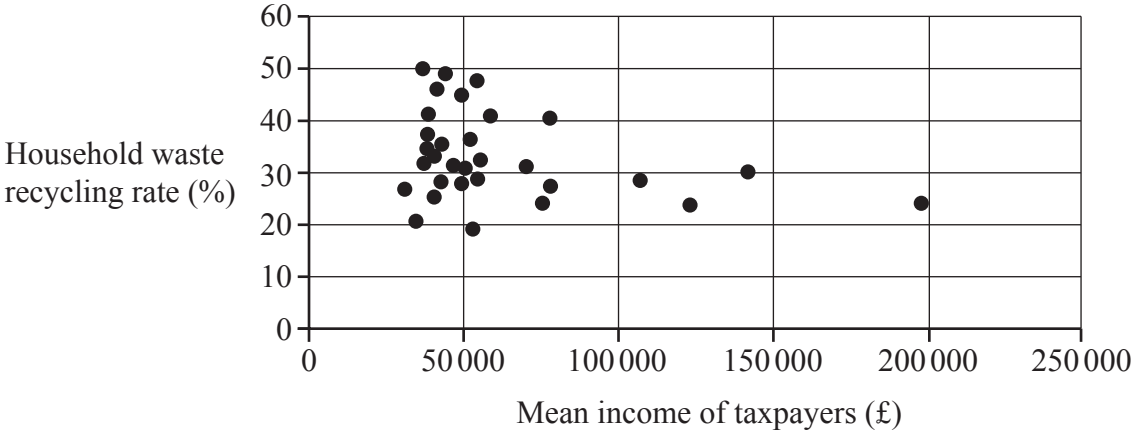
7 The pre-release material contains data on household waste recycling rates and mean income of taxpayers in different areas of England.

The scatter diagram shows the household waste recycling rates and mean income of taxpayers for different areas of London in 2020/21.



(a) Initially Umi thinks that the scatter diagram suggest that there is some negative correlation between household recycling rates and mean income of taxpayers.

Explain why Umi might believe this is the case. **[1]**

Having inspected the diagram more closely, Umi believes that there are **4** outliers.

(b) On the copy of the diagram in the **Printed Answer Booklet**, circle these four outliers. **[1]**

The table below shows further analysis of mean income of taxpayers and household waste recycling rates in 2020/21 in areas of London.

	Lower Quartile	Upper Quartile
Mean income of taxpayers (£)	40 350	64 300
Household waste recycling rate (%)	27.80	38.85

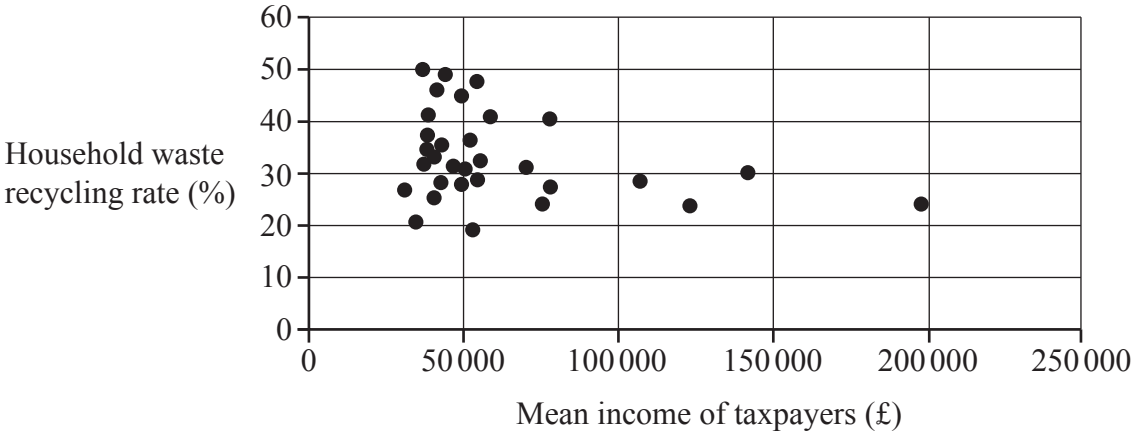
(c) Use a suitable calculation to explain whether or not this information supports Umi's statement. **[2]**

Umi removes all 4 points which they believe represent outliers and draws a new scatter diagram.

(d) Describe the correlation between household waste recycling rates and mean income of taxpayers in London suggested by Umi's **new** diagram. **[1]**

7(a)	

7(b)


Household waste recycling rate (%)
60
50
40
30
20
10
0
0
50 000
100 000
150 000
200 000
250 000
Mean income of taxpayers (£)

7(c)

7(d)